

COL1000: Lab 2

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: `<entryNumber>_q<quesNumber>.py`. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements.

Question 1: Vowels and Consonants Write a program that reads a single letter and checks:

- If it is a vowel (a/A, e/E, i/I, o/O, u/U), print `Vowel`.
- Otherwise, print `Consonant`.

Question 2: Series Sum Write a program that reads two positive integers k and n ($k \leq n$). Using a while loop, compute and print the sum of the cubes of all integers from k to n inclusive; that is:

$$k^3 + (k + 1)^3 + \cdots + n^3.$$

Question 3: Hollow square Write a program that reads two positive integers n and d from the user (assume $2d < n$), and prints a hollow square of side length n with border thickness d made of `*` characters, using a `while` loop along with string concatenation operators `*`, `+`.

For example, if $n = 7$, $d = 2$, the output is:

```
*****
*****
**    **
**    **
**    **
*****
*****
```

Question 4: Palindrome Write a program that reads a positive integer n and prints `True` if n is a palindrome, and `False` otherwise.

For example, if the input is 45654, then the output must be `True`. If the input is 4654, then the output must be `False`.

Note: You must not use list conversion, or built-in operations such as `reverse`, `join`, `string-slicing`, or `len`.